
Initial validation of DMD on sparse calcium-event data

Step-by-step implementation, diagnostics, controls, and gate decision

Frozen screening decision: FAIL

The native deconvolved arm predicts better than persistence, but the pilot fails known-system recovery, invariant-subspace reproducibility, tracking resolution, and the coordinated-mode null qualifier. Intensive sliding-window DMD/Grassmann tracking is not authorized from this checkpoint.

Project: `ca_imaging_network_analysis/DMD`
Report: Version 1.0
Accepted run: `20260716T233537+0900_ca0ddd9_3dc26e9ab8`
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1 Executive conclusion

Decision in one sentence

The intended 300-frame sliding-window DMD analysis is **not presently a defensible basis for tracking brain-state dynamics**: prediction exists, but the dynamic modes are neither recoverable under matched sparse observations nor reproducible/resolved well enough for Grassmann-manifold tracking.

This conclusion is deliberately narrower than “DMD does not work.” The implementation is numerically healthy, reproduces the direct precedent’s operator, and recovers continuously observed known systems. The failure occurs after the latent dynamics are observed as extremely sparse deconvolved events or calcium-like traces. In the empirical data, a heavily regularized DMD model often beats persistence, but independent per-PC autoregression explains almost all of that advantage. Neuronwise negative controls retain similar predictability.

The five frozen gates gave:

Gate	Result	Decisive observation
1. Geometry and numerical health	PASS	No illegal joins; 100% finite; no explosive forecasts.
2. Known-system recovery	FAIL	Classification 0.55 and median subspace overlap 0.430.
3. Empirical prediction	PASS	Conditional P-arm lower bootstrap bound versus persistence above zero in 4/4 fixed cells.
4. Empirical reproducibility	FAIL	0/4 cells reach $S_{\text{sub}} \geq 0.80$; matching failures occur.
5. Tracking resolution	FAIL	$R_{\text{track}} = 0.819$ and 0.854 , versus required < 0.5 .

No cell satisfied the additional requirement that full-DMD gain over diagonal AR be both positive and above the 95th percentile of the neuronwise circular-shift null. Therefore the report supports *forecastable marginal relaxation* at most, not coordinated or rotational population modes.

Recommended pause

Do not proceed automatically to all-recording sliding-window DMD or Germain-style Grassmann tracking. The user review should decide whether to stop this route or commission a new plan for a calcium-aware latent state-space model, coarser rate/count observations, or much longer windows. Any revision is a new protocol, not a reinterpretation of this failed screen.

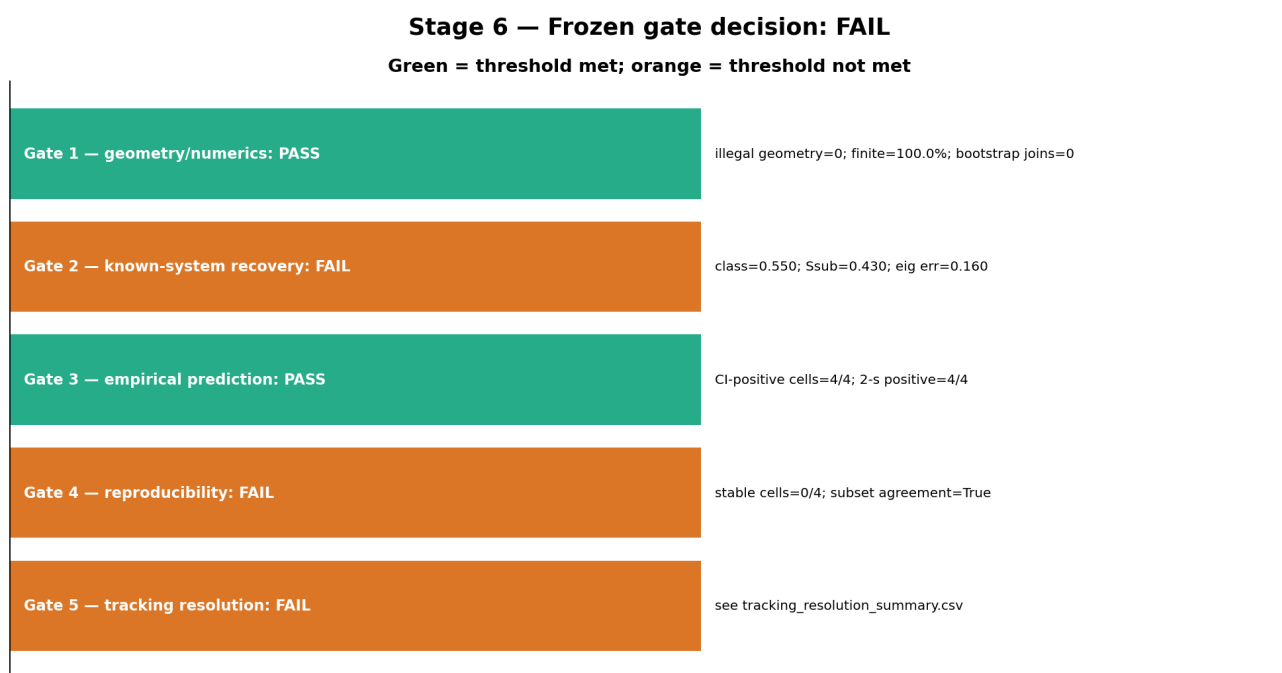


Figure 1: Final frozen-gate summary. Green marks the two thresholds that passed; orange marks the three that failed.

2 Scope and interpretation rules

The pilot tested whether DMD could be a credible measurement tool before asking whether awake, NREM, or anaesthesia differ. It did *not* estimate state effects. State labels were used only to construct legal constant-label windows and to summarize untouched results after development-only model selection.

The direct methodological anchor was the Pachitariu et al. analysis of neuronwise RMS-standardized deconvolved activity with lagged ridge DMD [1]. Standard DMD background follows Schmid [2] and Tu et al. [3]. The present pilot adds frozen recording-level PCA, contiguous held-out prediction, diagonal-AR baselines, known-system observation controls, block bootstrap, proper-subspace projectors, equal-neuron subsets, and neuronwise temporal nulls.

Three distinctions govern every interpretation:

1. Beating persistence is evidence of predictability, not necessarily of a multivariate population operator.
2. Beating diagonal AR is required for coordinated cross-PC dynamics.
3. A smooth-looking subspace trajectory is not trackable unless within-window estimation noise is materially smaller than ordinary between-window variation.

3 Audit trail and implementation revisions

3.1 Chronology

Intermediate results were inspected before the next expensive stage. Earlier runs were preserved rather than overwritten:

Run	Last completed stage	Purpose / reason superseded
20260716T224117. . .	Data	First figure geometry check; title/legend layout revised.
20260716T224223. . .	Preprocessing	PC-trace layout audit; figure spacing revised.
20260716T224405. . .	Forecast	First full predictive audit; exposed split, long-baseline, and provenance issues.
20260716T232145. . .	Nulls	Corrected implementation audit; exposed the need to require positive gain in the null qualifier.
20260716T233537+0900_ ca0ddd9_3dc26e9ab8	Decision	Final accepted, source-consistent run reported here.

3.2 Declared amendments

Four clarifications were frozen before the first empirical DMD fit: (i) require $q \geq 8$ for a $k = 4$ tracking subspace, (ii) use exact global maximin window placement with earliest ties, (iii) make development validation local 80/20 to match deployment, and (iv) add a seven-frame stored-smoothing edge sensitivity.

The stepwise audit then produced three explicit corrections before the accepted rerun:

1. Gate 2 failed, so downstream stages continued only under the configured label `diagnostic_completion_only_not_confirmatory_validation`.
2. The edge-trimmed sensitivity now preserves the original frame-240 train/test boundary: after retaining frames [7, 293), the trimmed-coordinate fitting stop is 233. The earlier implementation had inadvertently moved it.
3. The long-block diagonal AR ridge is selected independently within the nested development split. Fourteen of 20 arm-by-cell fits choose a ridge different from the full DMD model.

The final null qualifier also requires a *positive* full-over-diagonal gain, not merely a value above a negative null percentile. Incomplete eigengroup matching automatically makes a cell ineligible for Gates 4 and 5.

4 Data, windows, and leakage safeguards

4.1 Recording inventory

Recording	Neurons	Frames	Minutes	Segments	Role
example_data	1,000	9,350	20.37	1	precedent check
mouse02_sleep	6,574	18,700	40.74	2	awake / NREM
mouse07_ane	4,337	19,000	41.39	2	awake / anaesthesia

At 7.65 Hz, a 300-frame window lasts 39.22 s. Native-arm models fit the first 240 frames (31.37 s) and score the final 60 frames (7.84 s). The prespecified 1,500-frame diagnostic lasts 196.08 s and uses a nested 64/16/20 split.

Six 300-frame windows were selected in each recording-label cell by globally maximizing minimum start separation. Chronological positions 1, 3, and 5 are development; 2, 4, and 6 are untouched evaluation. No published “used-frame” mask was applied, so the early anaesthesia interval remains available. This is appropriate for a general dataset-use tutorial and avoids silently deleting transition-related frames.

Geometry audit

There are 24 deployment windows, 12 development windows, 12 untouched evaluation windows, and four long diagnostics. There are zero development– evaluation overlaps, zero state-label violations, and zero acquisition-boundary crossings. Snapshot pairs are formed separately inside each legal window or bootstrap block. Thirteen deployment windows touch the approximate support of the stored 15-frame smoother; this motivates the explicit S_z/S_c sensitivity, but it does not affect the unsmoothed P arm.

5 Stage 1: candidate observations and frozen PCA

Five observation arms were fit from development data only:

Arm	Signal / scaling	Purpose
P	deconvolved; $\text{RMS}+10^{-3}$	Pachitariu-aligned anchor
B4	four-frame event mass; $\text{RMS}+10^{-3}$	coarser mean-rate/count-like arm
S_z	stored smoothed; $\text{RMS}+10^{-3}$	code-matched smoothing control
S_c	stored smoothed; center only	scaling sensitivity
F	$\Delta F/F$; $\text{RMS}+10^{-3}$	continuous calcium control

Recording	Arm	Exact zeros	Eligible neurons	Lag-1 autocorrelation
mouse02 sleep	P	98.91%	6,553	0.125
	B4	96.69%	6,553	0.249
	Sz	90.38%	6,556	0.979
	Sc	90.38%	6,556	0.981
	F	0%	6,574	0.699
mouse07 anaesthesia	P	99.51%	4,193	0.046
	B4	98.31%	4,193	0.174
	Sz	94.94%	4,200	0.978
	Sc	94.94%	4,200	0.979
	F	0%	4,337	0.697

The event observations are therefore more discontinuous than a simple binary description suggests: they are almost always exactly zero, with sparse continuous-amplitude events. Stored smoothing and $\Delta F/F$ create strong memory, so good persistence forecasts from those arms are not by themselves evidence for dynamic modes.

All ten recording–arm PCA bases passed the orthogonality check; the maximum Frobenius error was 9.88×10^{-15} . No nonfinite neuron was found. The heatmaps below include *all* eligible neurons; no 1,000-neuron display cap or row sampling is used.

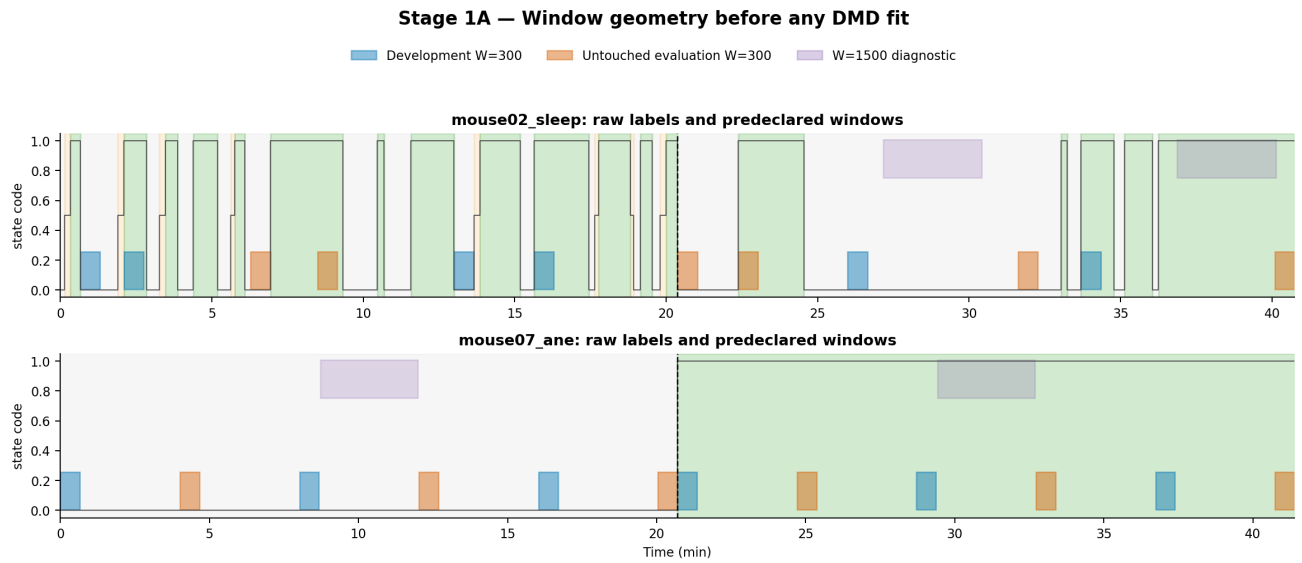


Figure 2: Raw state labels, exact development/evaluation windows, acquisition boundaries, and prespecified long blocks before any DMD fit.

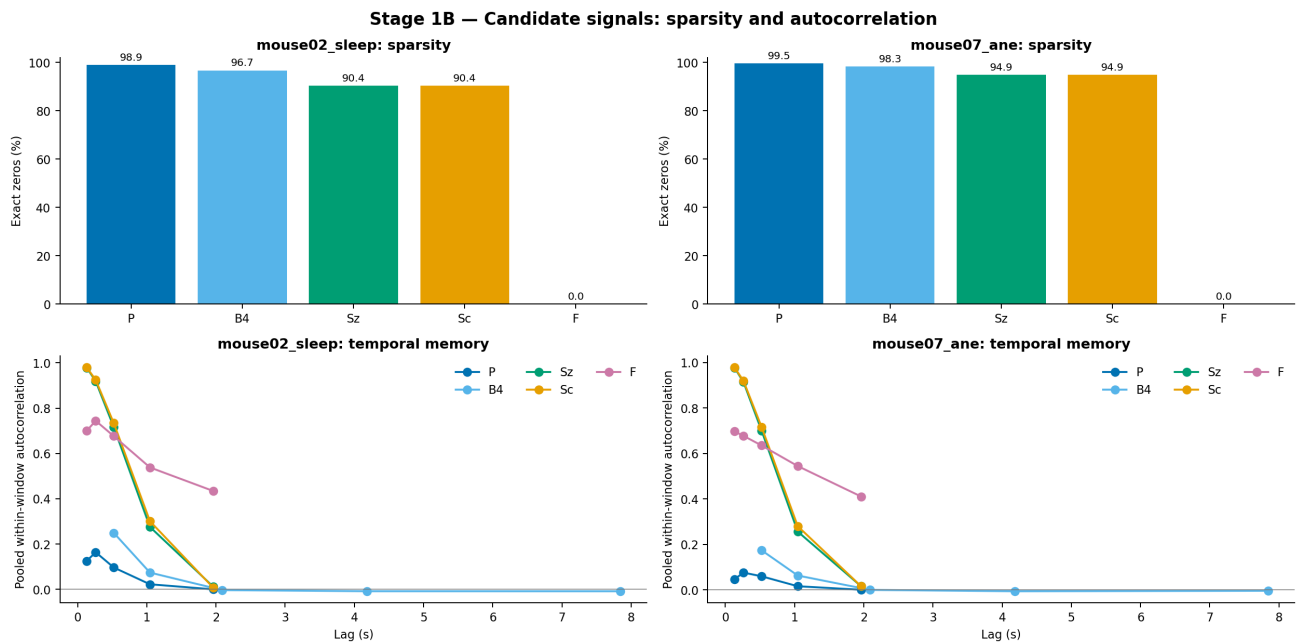


Figure 3: Sparsity and pooled within-window autocorrelation for all five candidate observation arms.

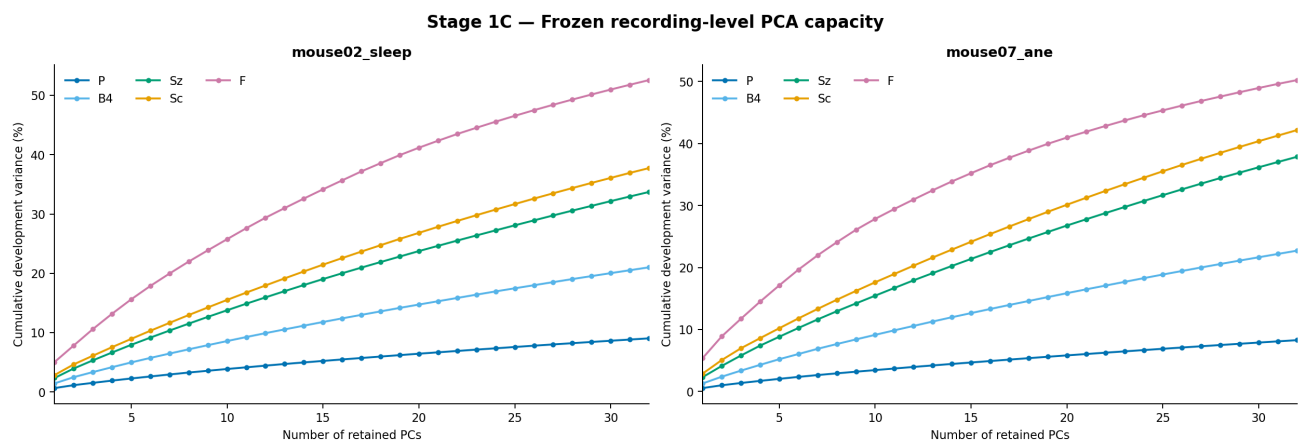


Figure 4: Cumulative development variance in the frozen recording-level PCA bases. The same basis is used for every state and evaluation window of a recording/arm.

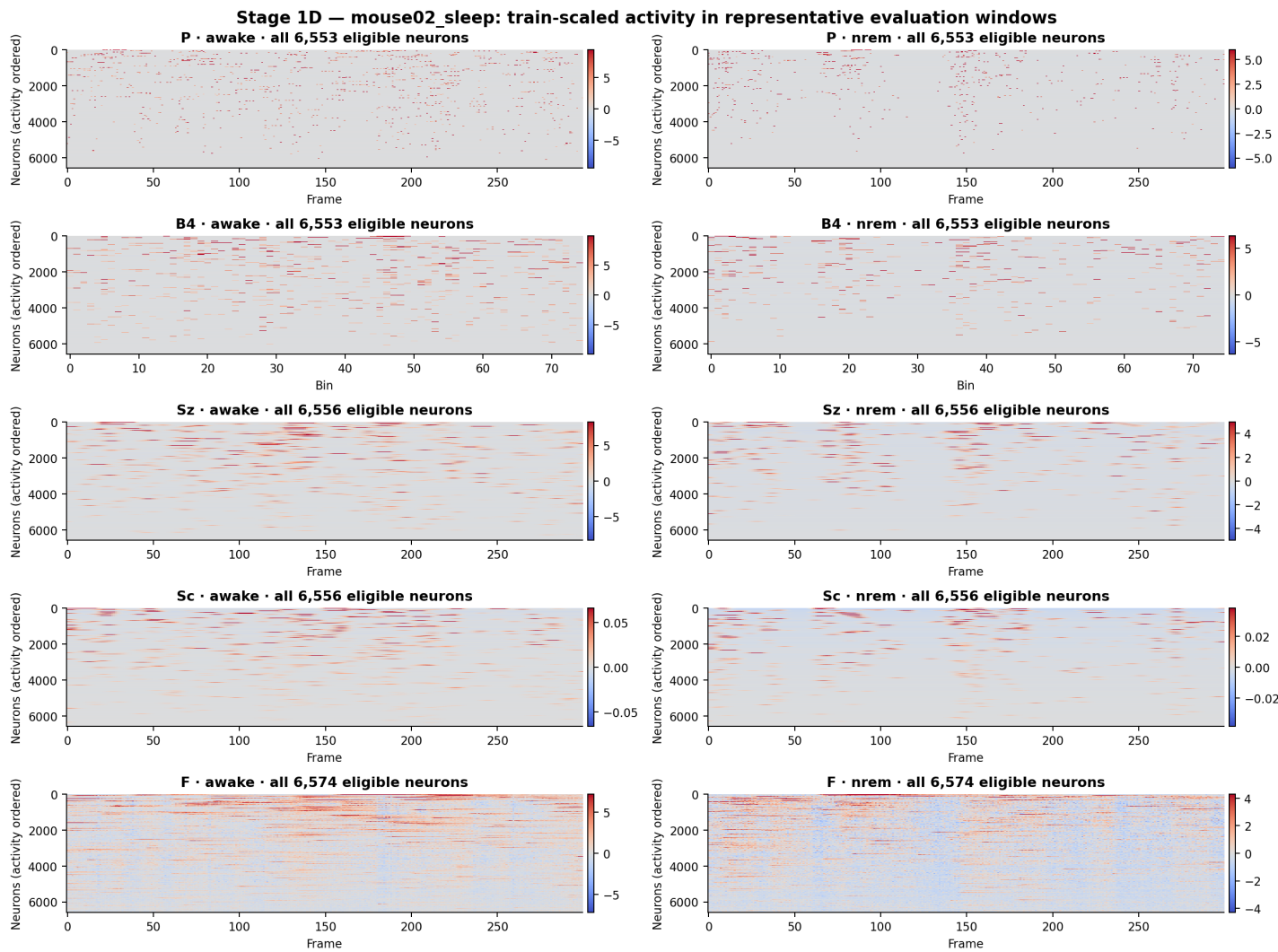


Figure 5: Mouse02 representative untouched windows: every eligible neuron, activity ordered separately within each panel.

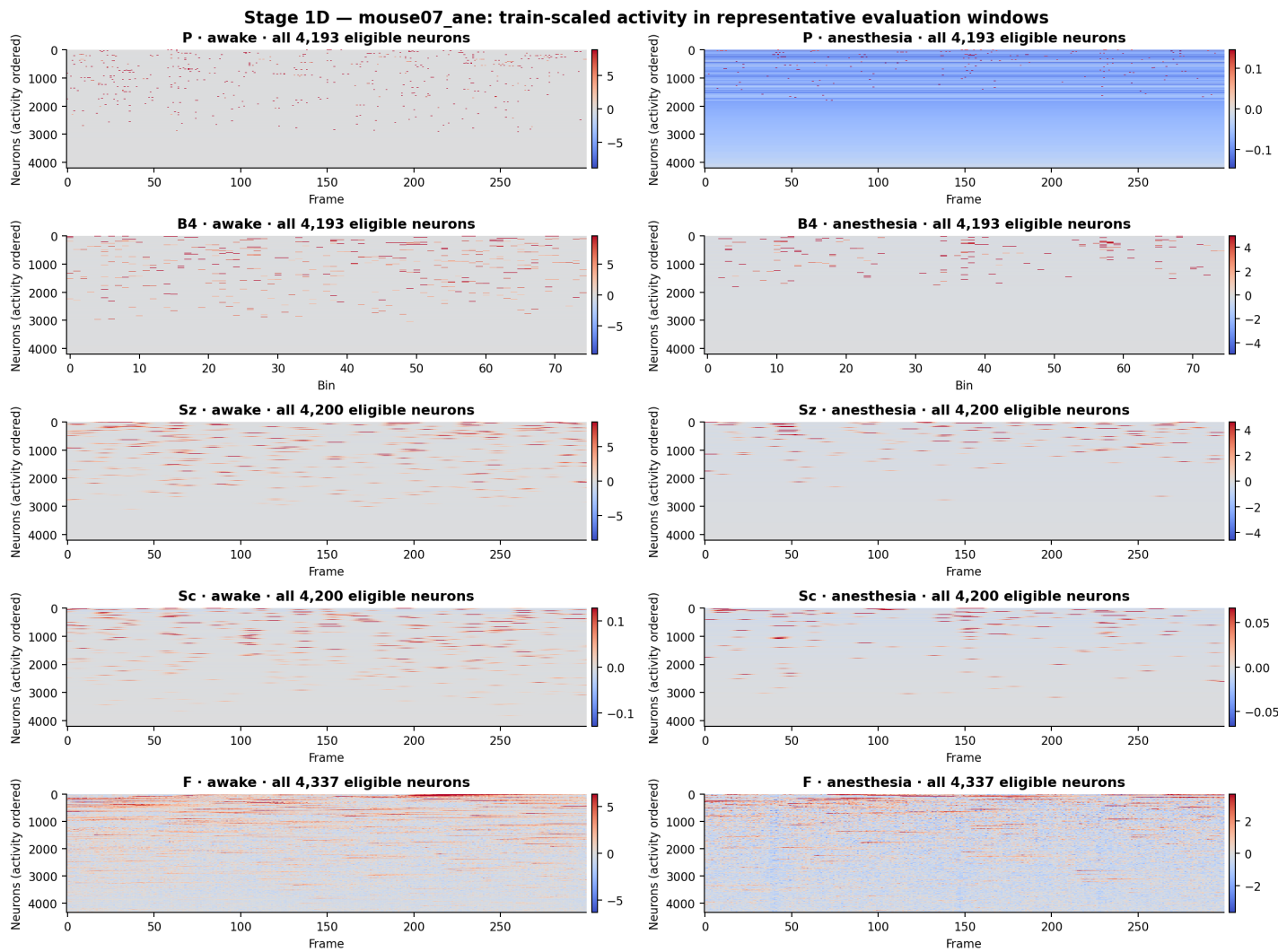


Figure 6: Mouse07 representative untouched windows: every eligible neuron, including awake and anaesthesia.

6 Stage 2A: direct-precedent implementation check

The precedent smoke test retains all 1,000 standardized neuron dimensions in `example_data`, uses a two-frame lag and raw ridge $\alpha = 0.01$, and fits 9,348 legal snapshot pairs. The stable thin-SVD operator matches the code-style Gram solve with relative Frobenius error 1.52×10^{-14} . Its spectral radius is 0.658. The implementation is thus algebraically faithful to the anchor calculation.

This is not a biological validation: the in-sample normalized residual is 0.897, and the eigenspectrum alone does not demonstrate held-out utility. The audit also retains the precedent discrepancy between the released $|\lambda| > 0.25$ code filter (577 modes) and a real-part threshold described in the caption (421 modes).

7 Stage 2B: known-system recovery

Two six-dimensional stable systems—real relaxation and one containing a known rotational pair—were observed through four pipelines over 20 deterministic seeds each. Continuous observations provide a software upper bound; event, smoothed-event, and calcium observations test identifiability under the present measurement problem.

System	Observation	Class accuracy	Subspace overlap	Eigenvalue error	Skill vs persistence
Real	continuous	1.00	0.963	0.027	0.185
Real	event	0.40	0.362	0.972	0.456
Real	smoothed event	1.00	0.443	0.079	0.130
Real	calcium	1.00	0.512	0.102	0.313
Rotational	continuous	1.00	0.998	0.017	0.656
Rotational	event	0.90	0.367	0.949	0.484
Rotational	smoothed event	0.00	0.391	0.077	0.143
Rotational	calcium	0.00	0.674	0.227	0.351

Across the transformed observations, classification accuracy is 0.55 and median subspace overlap is 0.430, below the frozen 0.80 and 0.75 thresholds. Eigenvalue error (0.160) and forecast skill (0.324) pass their components. The combination is scientifically important: smoothing/calcium can make a signal easy to forecast while erasing the distinction between real and rotational dynamics. Forecasting cannot substitute for mode recovery.

The aggregate eigenvalue error also conceals an observation-specific failure: event observations have median error 0.954, whereas the smoother observation classes pull the transformed-observation aggregate below threshold. The valid conclusion is therefore that this observe-PCA-ridge-DMD pipeline does not preserve identifiable modes through the tested sparse/calcium-like observation maps. It is not evidence that DMD is numerically broken or that no event-data DMD method could work.

Gate 2 failure and diagnostic override

Gate 2 failed before empirical forecasting. Downstream work was nevertheless completed because the user requested a full smoke-test diagnosis. Every later result is labelled diagnostic and cannot reverse this frozen failure.

8 Stage 3A: anonymous development selection

For rank q , lag ℓ , PC score z_t , and relative ridge η , the fitted operator is

$$z_{t+\ell} = Bz_t + \varepsilon_t, \quad \hat{B} = YX^\top (XX^\top + \alpha I)^{-1}, \quad \alpha = \eta \frac{\text{tr}(XX^\top)}{q}. \quad (1)$$

It is computed with a thin SVD rather than an explicit inverse. Development selection fits the first 80% and scores the last 20% of each of six anonymous development windows per recording. Labels do not enter the ranking.

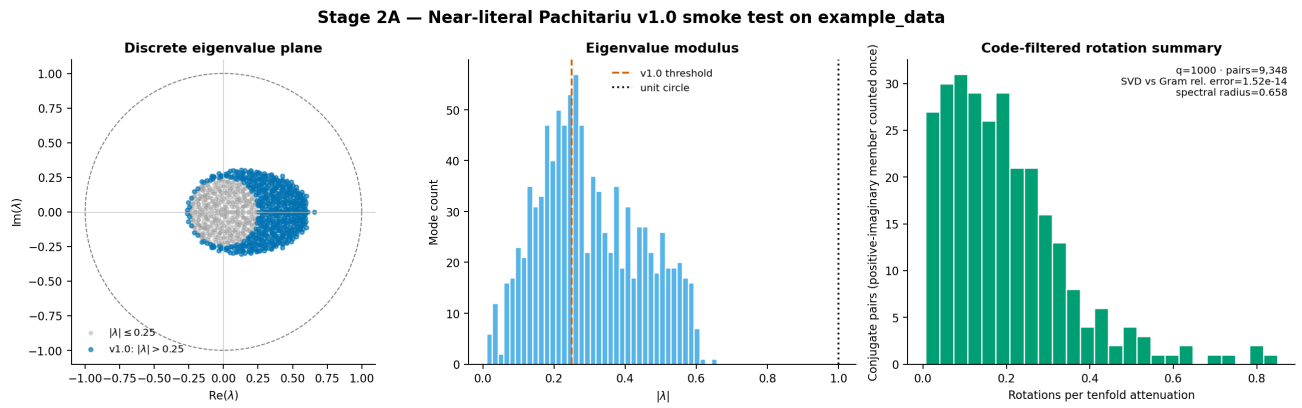


Figure 7: Near-literal precedent check. Rotation summaries count the positive-imaginary member of each conjugate pair once.

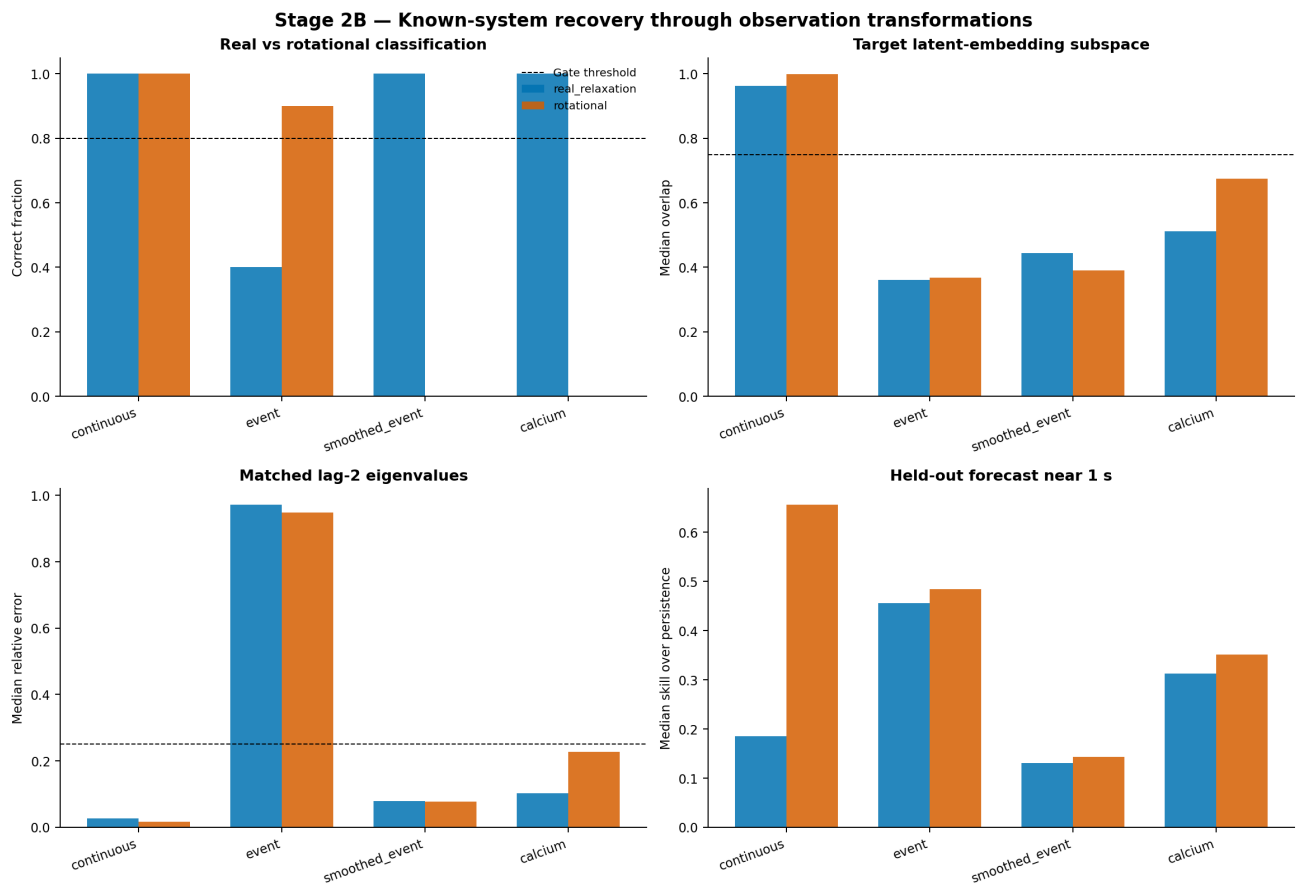


Figure 8: Known-system recovery at four observation levels. Dashed lines show frozen thresholds. Continuous observations validate the software; transformed observations expose the identifiability failure.

Median development scores selected P globally (0.389), followed by B4 (0.293), F (0.214), Sc (0.151), and Sz (0.128). The chosen P configurations are:

Recording	Role	q	ℓ	η	diagonal η
mouse02	prediction	4	4	1	10^{-2}
mouse02	tracking	8	2	10^{-1}	10^{-4}
mouse07	prediction/tracking	16	8	1	1

The mouse02 rank-4 winner is prediction-only: a $k = 4$ projector in $q = 4$ would be the identity. Tracking therefore uses the prospectively selected $q = 8$ configuration. Several winners lie at grid boundaries, so the selected ranks and lags should not be interpreted as precisely localized biological scales.

9 Stage 3B–G: untouched empirical forecasts and spectra

9.1 Prediction against persistence and diagonal AR

The P arm’s median untouched near-1 s skills are:

Cell	vs persistence	vs diagonal AR	equal-PC vs persistence
mouse02 awake	0.449	−0.008	0.494
mouse02 NREM	0.185	−0.007	0.044
mouse07 anaesthesia	0.455	0.082	0.440
mouse07 awake	0.516	0.009	0.511

At near 2 s, P remains positive versus persistence in all cells (0.224–0.475), but full-over-diagonal gains remain −0.068, −0.001, 0.004, and 0.089. All primary operators, eigenvalues, SSE values, and forecasts are finite; the explosive fraction is zero and the maximum spectral radius is below one.

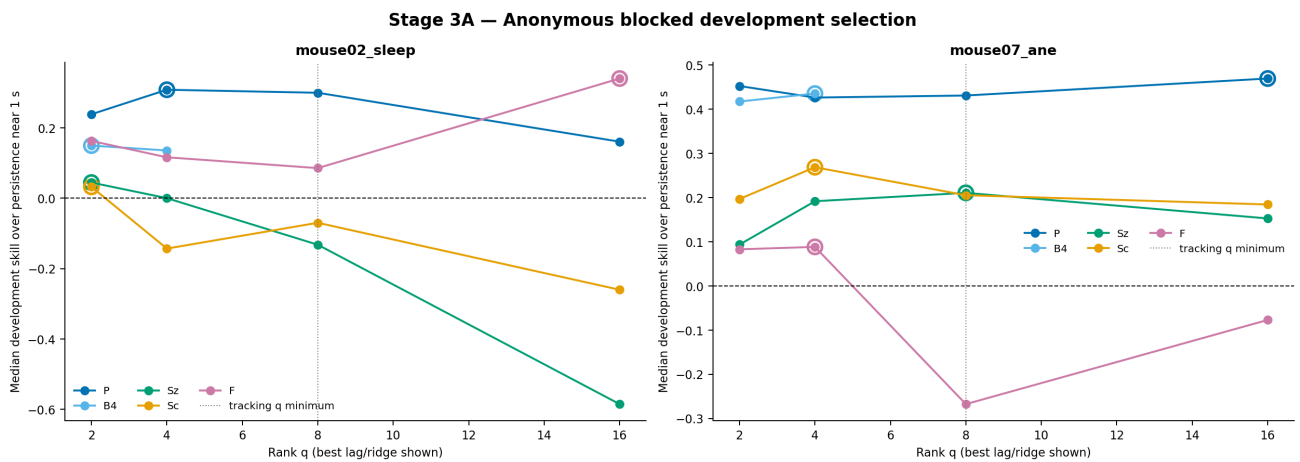


Figure 9: Anonymous blocked development selection. Hollow circles identify prediction winners; the vertical line marks the minimum rank eligible for a proper $k = 4$ tracking subspace.

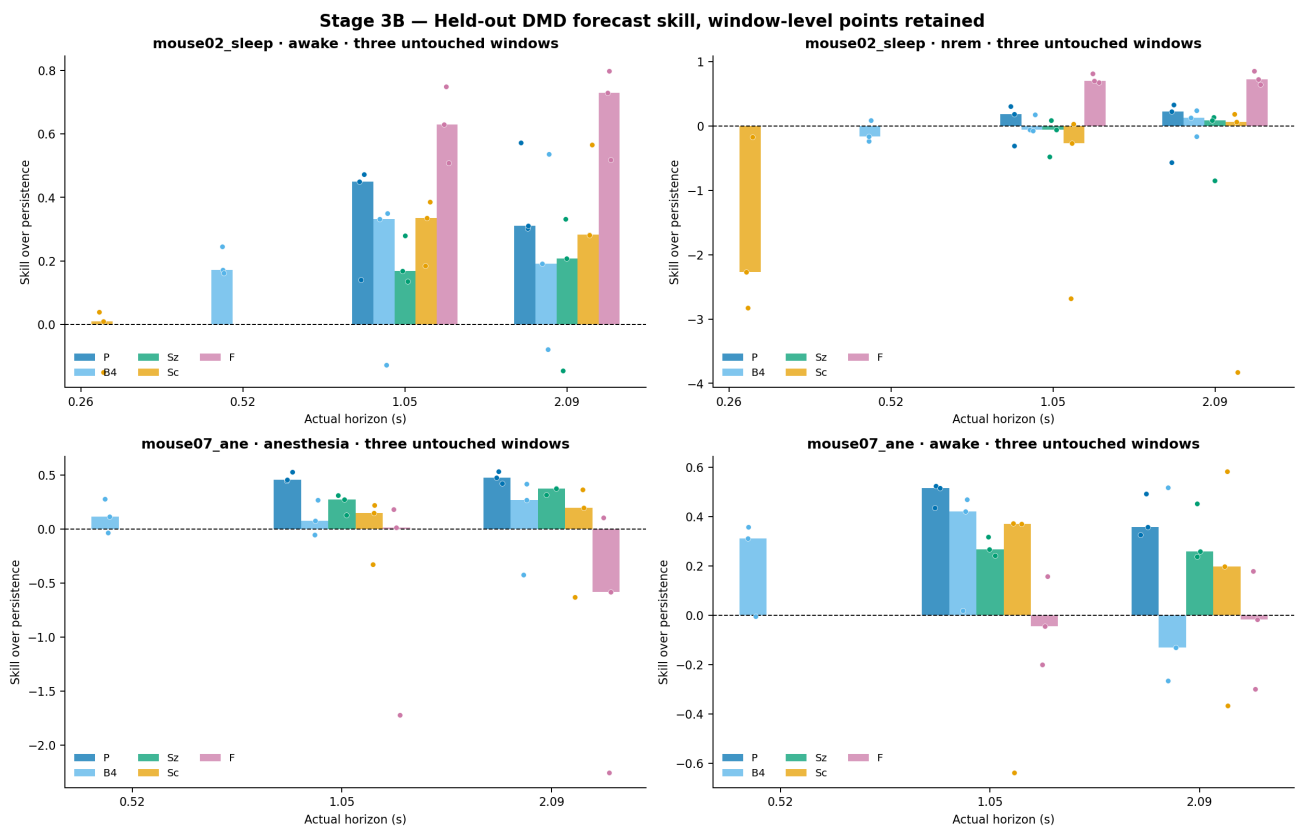


Figure 10: Held-out skill over persistence at all representable physical horizons. Bars are medians and dots retain the three untouched window values.

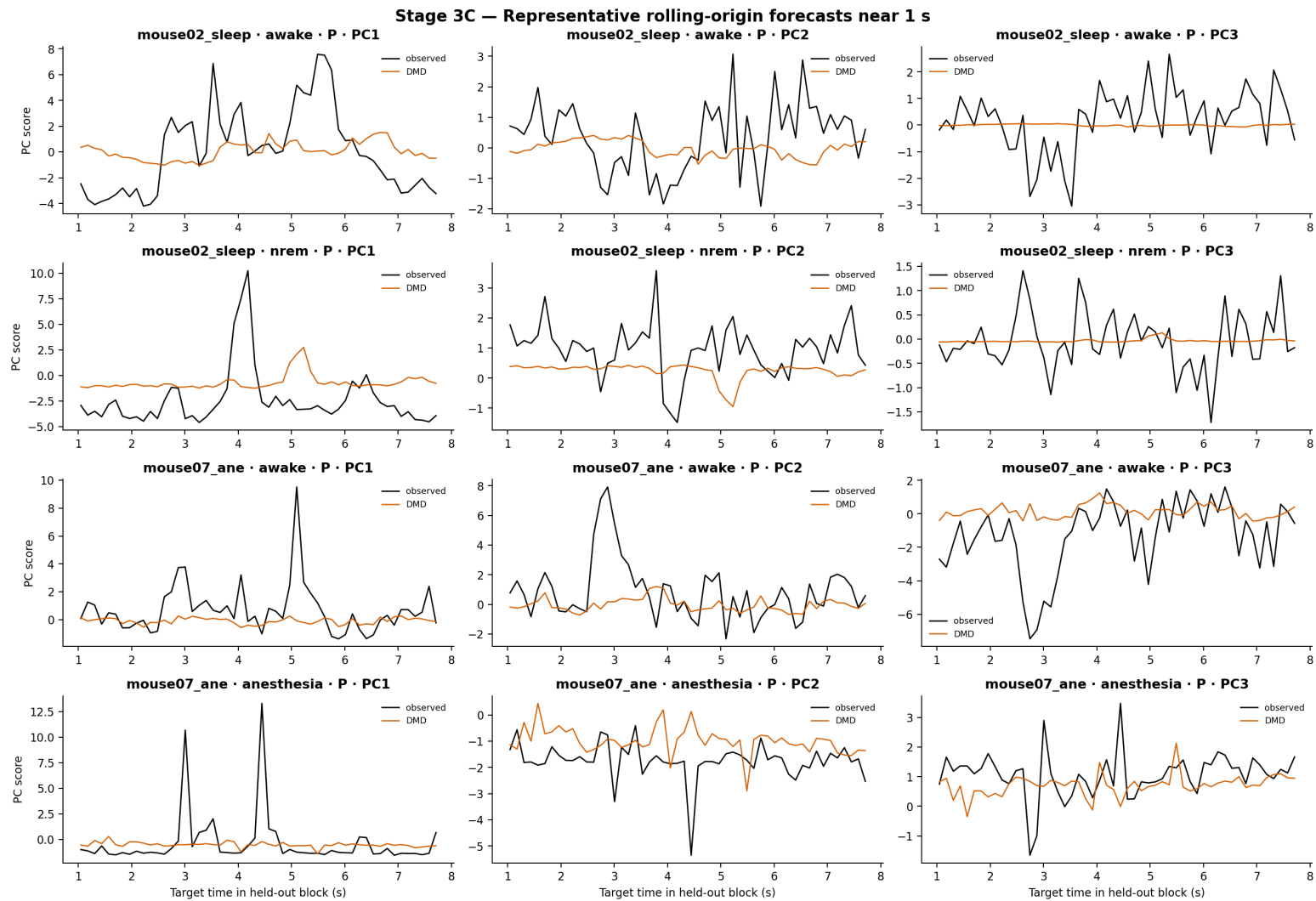


Figure 11: Chronologically earliest untouched-window P forecasts near 1 s. Strong shrinkage toward a near-flat trajectory can beat sparse-signal persistence without reproducing event peaks.

What the predictive pass means

P meets the frozen conditional persistence criterion, but this is mostly not a multivariate DMD effect. Persistence carries a sparse event-like PC state forward; a regularized model that shrinks toward zero can outperform it. The diagonal AR result and the representative trajectories forbid interpreting Gate 3 as evidence for coordinated modes. Relative to the training-mean predictor, the cell-median near-1 s skill is positive only for mouse02 awake (0.130) and is negative for mouse02 NREM (−0.040), mouse07 anaesthesia (−0.100), and mouse07 awake (−0.086). Thus Gate 3 is a conditional estimator-perturbation pass against persistence in these fixed windows, not general predictive validity.

9.2 Discrete spectra

All selected predictive spectra are plotted directly in the discrete plane. The P arm has zero interpretable held-out rotational pairs after requiring stability, modulus > 0.25 , and at least three cycles in the fitting window. Complex spectra in other arms, especially $\Delta F/F$, are not sufficient to override their development ranking, baseline performance, or the known-system failure.

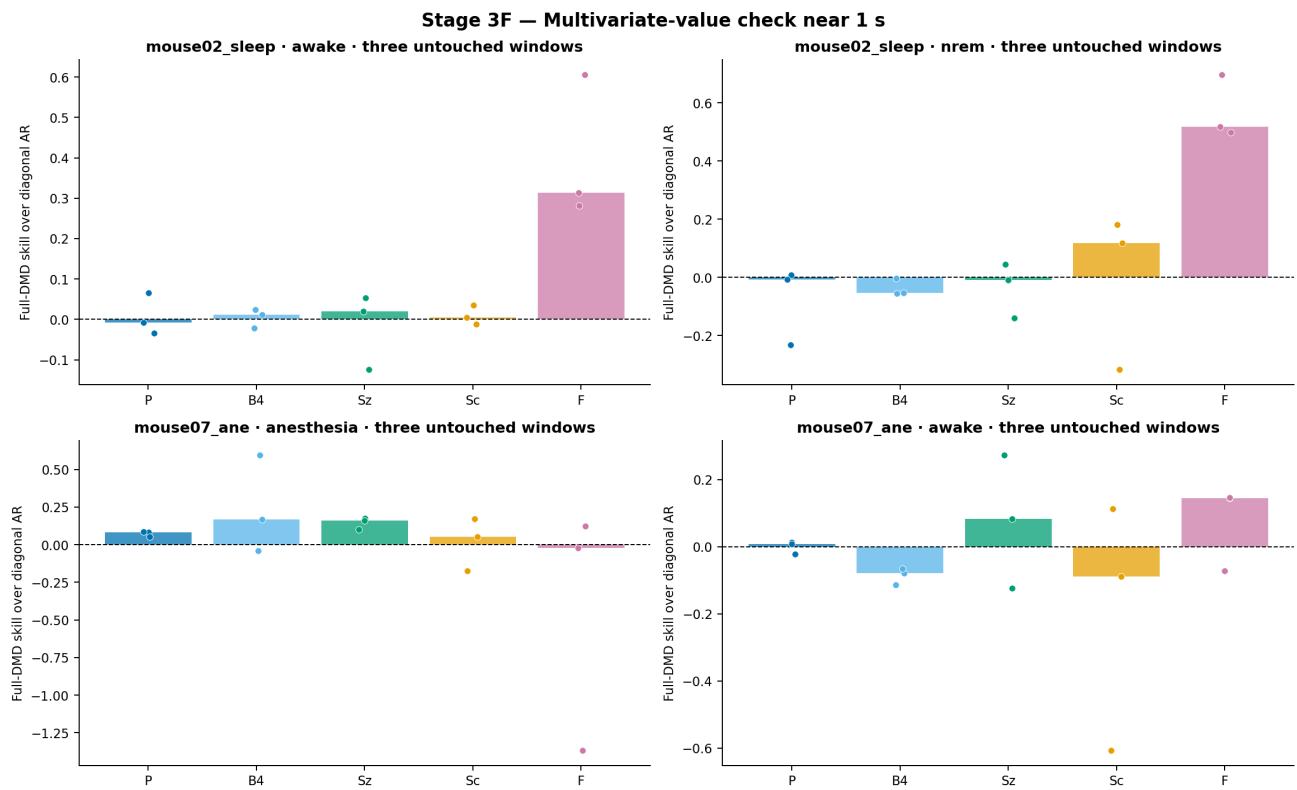


Figure 12: Full-DMD skill over the independently tuned diagonal-AR baseline. The P-arm bars are near zero in three cells and modestly positive only in anaesthesia.

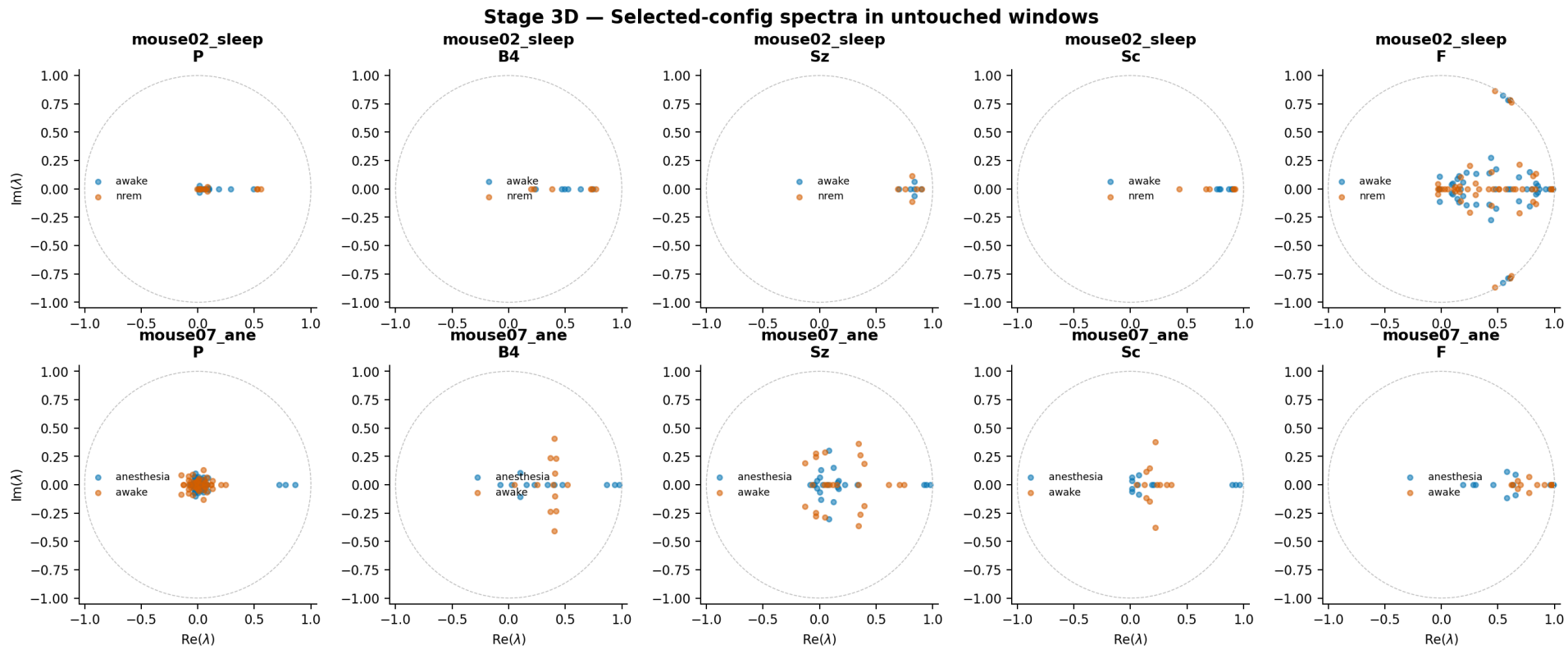


Figure 13: Selected predictive-configuration spectra in all untouched windows. The unit circle is shown; points are discrete-lag eigenvalues.

9.3 Long-block diagnostic

The prespecified long blocks improve P’s persistence skill most strongly in mouse02 NREM, from 0.185 to 0.443. However, these fits reselect larger cell-specific models (often $q = 32$) and therefore combine more temporal support with a larger search space. Their full-over-diagonal gains remain only 0.009 to 0.066 after independent diagonal-ridge tuning.

Cell	W=300 skill	W=1500 skill	Long q	Long ℓ	Long vs diagonal
mouse02 awake	0.449	0.482	32	8	0.012
mouse02 NREM	0.185	0.443	32	4	0.023
mouse07 anaesthesia	0.455	0.471	32	4	0.066
mouse07 awake	0.516	0.450	16	8	0.009

9.4 Stored-smoothing boundary sensitivity

After trimming seven frames at both edges while preserving the original split, median near-1 s persistence skill changes as follows:

Cell	Sz primary	Sz trimmed	Sc primary	Sc trimmed
mouse02 awake	0.168	0.160	0.335	0.336
mouse02 NREM	−0.058	0.076	−0.269	0.121
mouse07 anaesthesia	0.271	0.263	0.150	0.177
mouse07 awake	0.268	0.308	0.370	0.340

The sign reversals in mouse02 NREM make stored-smoothed conclusions boundary-sensitive. The full-range plot retains severe individual-window failures rather than clipping them; the table supplies the robust cell medians.

10 Stage 4: bootstrap prediction and mode reproducibility

10.1 Moving-block forecast uncertainty

Each evaluation-window fitting segment was resampled 100 times in moving blocks of at least 15 native frames. Pairs were created only inside blocks. Repetition b was synchronized across the three evaluation windows, their cell median was taken, and percentile intervals were then computed. These intervals quantify operator-estimation uncertainty for the fixed selected windows; they are not population-level confidence intervals over animals. They condition on the frozen PCA basis, hyperparameters, windows, and test tails and do not include test-origin, window-selection, neuron-sampling, preprocessing-choice, or model-selection uncertainty.

Cell	1-s bootstrap median	95% interval	full-over-diagonal 95% interval
mouse02 awake	0.450	[0.445, 0.452]	[−0.021, 0.001]
mouse02 NREM	0.183	[0.169, 0.196]	[−0.033, 0.026]
mouse07 anaesthesia	0.452	[0.443, 0.460]	[0.035, 0.091]
mouse07 awake	0.510	[0.497, 0.519]	[−0.030, 0.011]

Thus the conditional interval excludes zero versus persistence for all four fixed cells, while only anaesthesia has a positive full-over-diagonal interval.

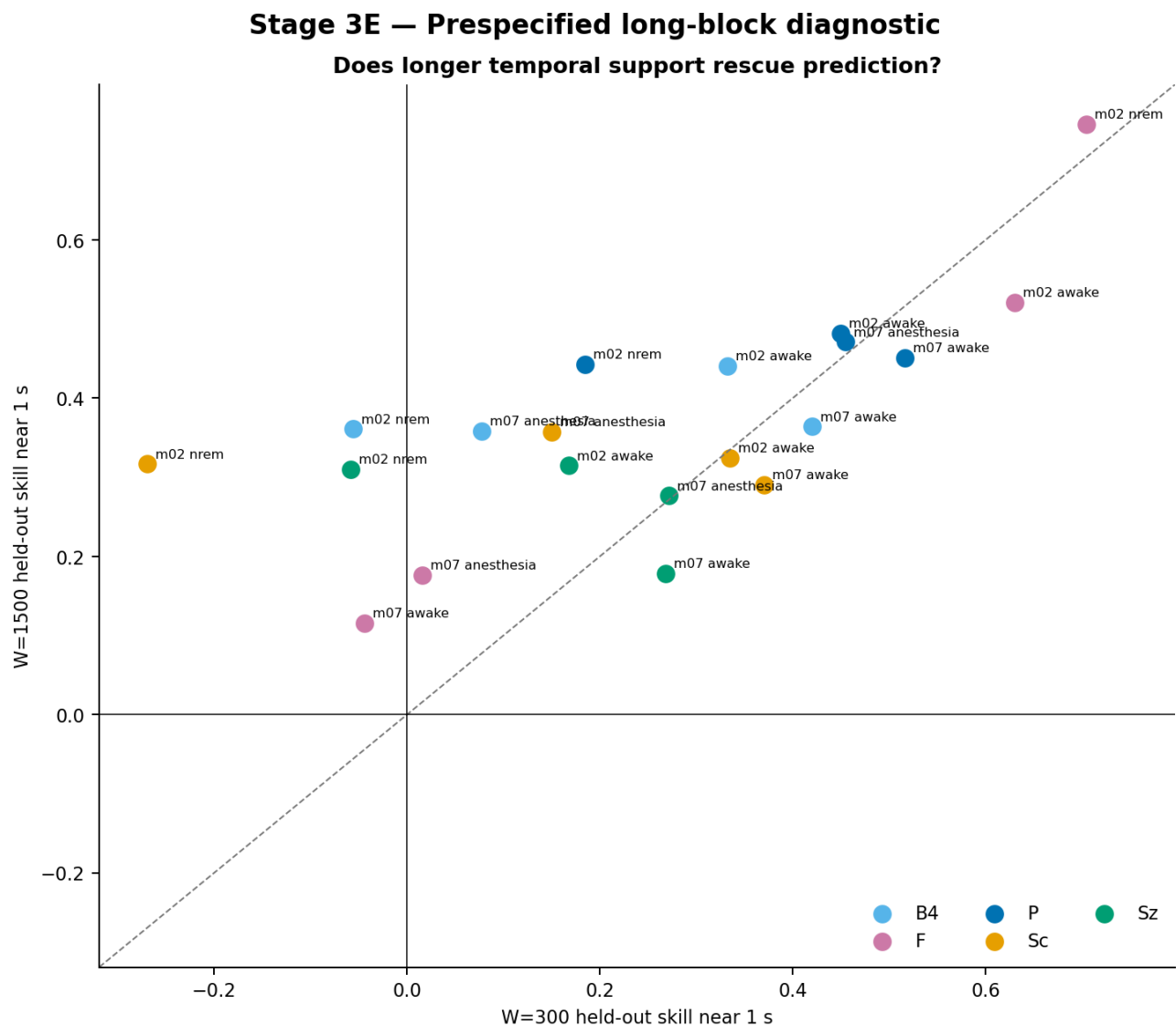


Figure 14: Cell-level comparison of 300-frame and separately selected 1,500-frame models. Points above the diagonal improve with the longer diagnostic; this is not a pure window-length contrast.

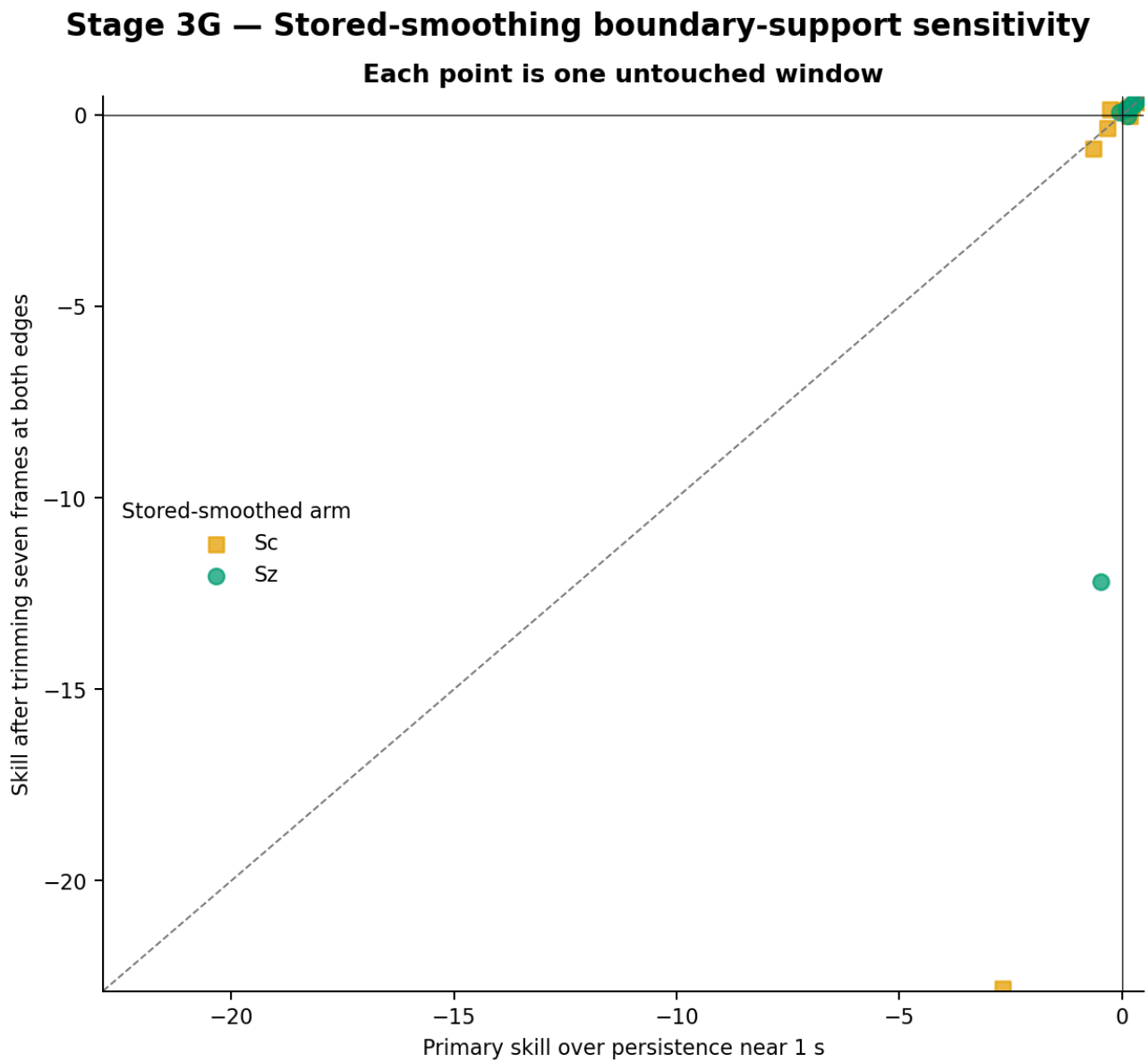


Figure 15: Primary versus fixed-split edge-trimmed skill for every untouched Sz/Sc window. Extreme negative values are deliberately retained.

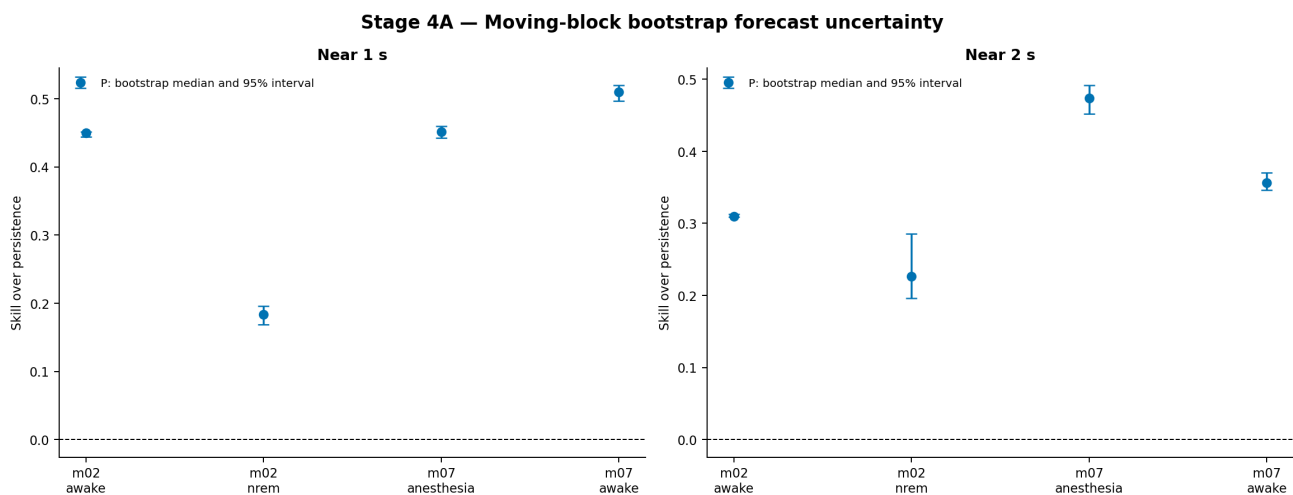


Figure 16: Synchronized moving-block intervals for P-arm skill over persistence at the near-1 s and near-2 s gates.

10.2 Fixed-reference invariant subspaces

For each recording, a development-only pooled operator defines $k = 4$ real invariant dimensions ranked by mean modal-coefficient energy. Conjugate pairs are never split. Evaluation and bootstrap eigengroups are matched to these fixed groups; no evaluation fit is allowed to re-rank a convenient subspace.

This frozen target itself requires caution. Modal-coefficient energy is based on the pseudoinverse of the eigenvector matrix and can be ill-conditioned or non-additive for nonnormal/repeated modes. Mouse07’s selected target comprises two complex pairs with $|\lambda| \simeq 0.077$ and 0.196 , both below the protocol’s $|\lambda| > 0.25$ interpretability cutoff and rapidly attenuated at the selected eight-frame lag. Mouse02 also includes modes at 0.242 and 0.283 . Low stability therefore characterizes this prespecified tracking object; it is not proof that no other dynamical representation exists.

Projector similarity and distance are

$$S_{\text{sub}} = \frac{1}{k} \text{tr}(PP^{(b)}), \quad d_P = \sqrt{\max(0, 1 - S_{\text{sub}})}. \quad (2)$$

Cell	Median S_{sub}	Within d_P	Match fraction	Min. windows/rep	Gate eligible
mouse02 awake	0.535	0.549	0.670	1	no
mouse02 NREM	0.500	0.570	0.813	1	no
mouse07 anaesthesia	0.238	0.754	1.000	3	yes
mouse07 awake	0.262	0.753	1.000	3	yes

Mouse02’s development reference selects four real one-dimensional groups, but some evaluation/bootstrap fits resolve complex pairs instead. The dimension-compatible matcher correctly refuses to coerce them. There are 157 recorded original/bootstrap matching failures. The two fully eligible mouse07 cells remain far below 0.80, so no cell passes the reproducibility threshold. Reported summaries condition on successful matches and are mildly optimistic: the assignment cost itself includes a 0.25-weighted version of the reported subspace-overlap endpoint.

10.3 Tracking resolution

The future tracking criterion compares within-window estimation distance with ordinary same-label between-window distance:

$$R_{\text{track}} = \frac{\text{median}(d_P^{\text{bootstrap}})}{\text{median}(d_P^{\text{between}})}. \quad (3)$$

Mouse02 has $R_{\text{track}} = 0.819$ but is ineligible because only four of six original evaluation projectors match. Mouse07 is eligible and has $R_{\text{track}} = 0.854$. Both exceed the 0.5 ceiling. Estimation noise is nearly as large as ordinary window-to-window change; a visually smooth Grassmann path would not be trustworthy. The between-window distances (approximately 0.685 and 0.879) are also close to independent-random-subspace expectations for the respective (q, k) dimensions (approximately 0.707 and 0.866).

10.4 Two disjoint neuron subsets

Two deterministic, disjoint 2,000-neuron subsets were independently reduced to the selected tracking rank. Their majority qualitative classes agree in all cells: resolved rotation for mouse02 awake and no resolved rotation for the other three cells. Agreement is useful but insufficient: it does not fix low reference similarity or poor tracking resolution.

11 Stage 5: empirical negative controls

For every evaluation window, 100 neuronwise circular shifts were generated inside the same acquisition-split, constant-label bout, with at least a 15-frame phase displacement. They preserve each neuron’s empirical

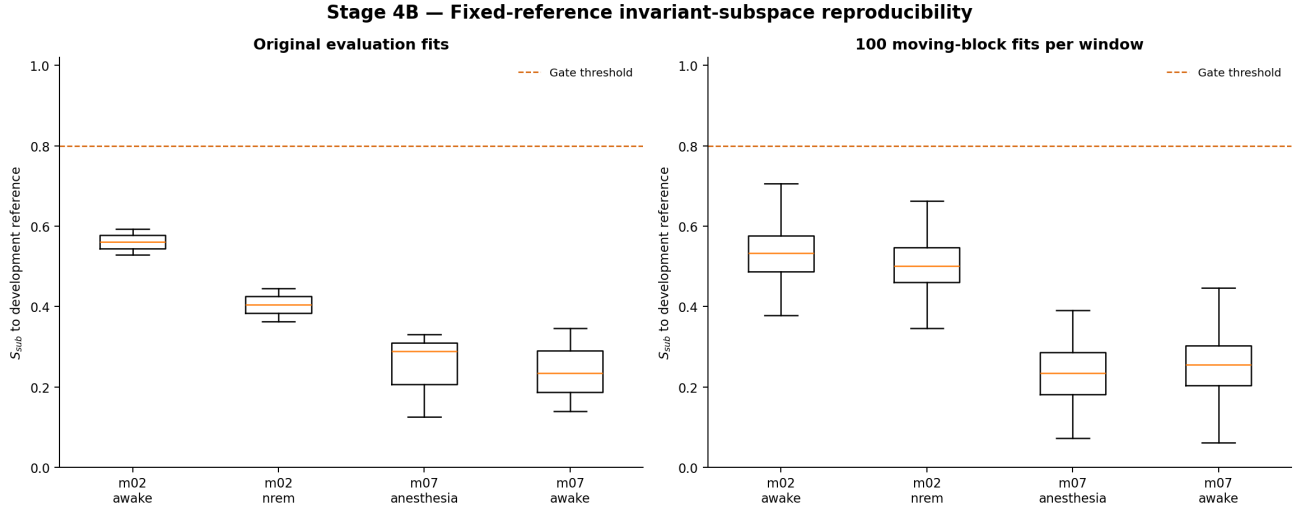


Figure 17: Similarity to the fixed development reference for original evaluation fits and moving-block fits. Unmatched fits are recorded as failures and excluded from boxes, making the table’s eligibility columns essential.

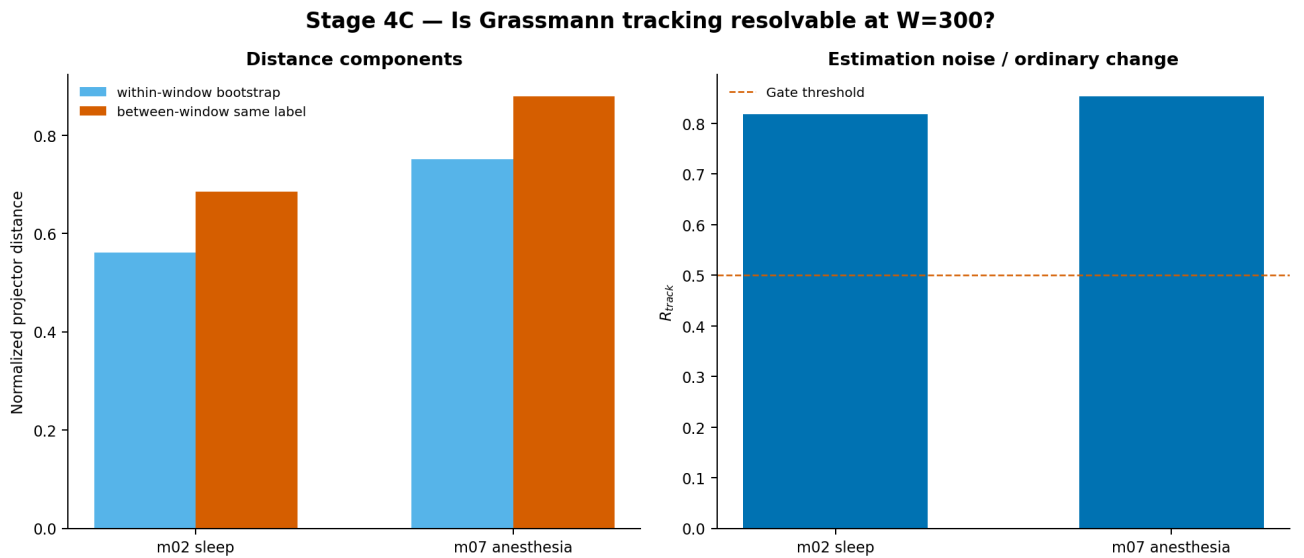


Figure 18: Tracking-resolution components and ratios. The dashed line is the frozen $R_{\text{track}} < 0.5$ threshold.

Stage 4D — Two deterministic disjoint 2,000-neuron subsets

Majority class across three untouched windows

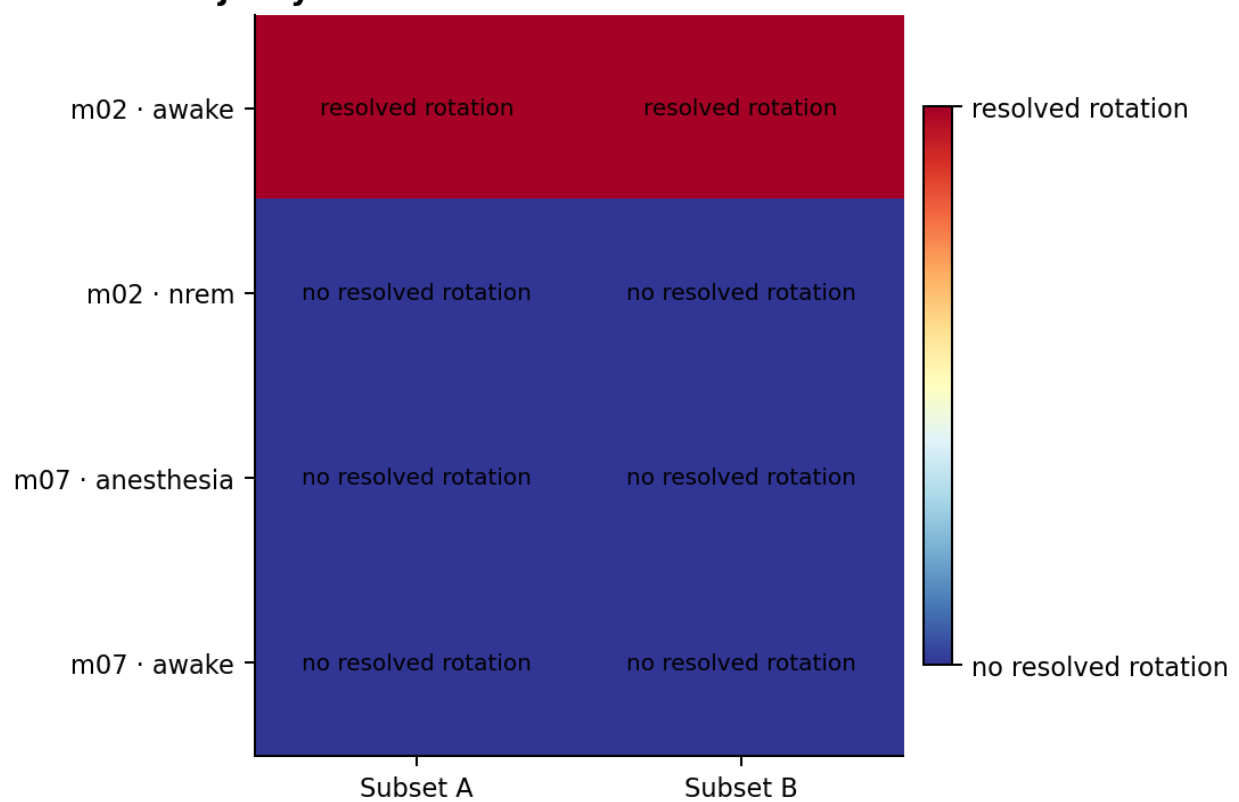


Figure 19: Qualitative spectral class in two equal-size disjoint neuron subsets.

marginal and autocorrelation approximately while disrupting synchrony. A second set of 100 stationary controls draws independently from each neuron’s within-bout empirical marginal. Both pass through the same frozen scaling, PCA, selected predictive model, tracking model, and scoring pipeline.

Circular shifts remain predictively useful versus persistence: their median skills are 0.320, 0.366, 0.456, and 0.457 across the four cells. Stationary-IID controls are similarly positive (0.331, 0.370, 0.467, and 0.468). This directly demonstrates that the empirical persistence result does not require preserved population synchrony or temporal dynamics; on a 99.2%-zero signal, shrinkage toward equilibrium can by itself beat persistence.

Cell	Observed full/diag	Circular median	Circular 95th	Positive?	Qualifier
mouse02 awake	−0.008	−0.089	−0.028	no	fail
mouse02 NREM	−0.007	−0.037	−0.006	no	fail
mouse07 anaesthesia	0.082	0.188	0.235	yes	fail
mouse07 awake	0.009	0.023	0.040	yes	fail

Mouse02 awake is above a negative null percentile, but its observed gain is itself negative and therefore cannot qualify. In the two positive cells, the observed value is below the circular 95th percentile. Zero of four cells pass.

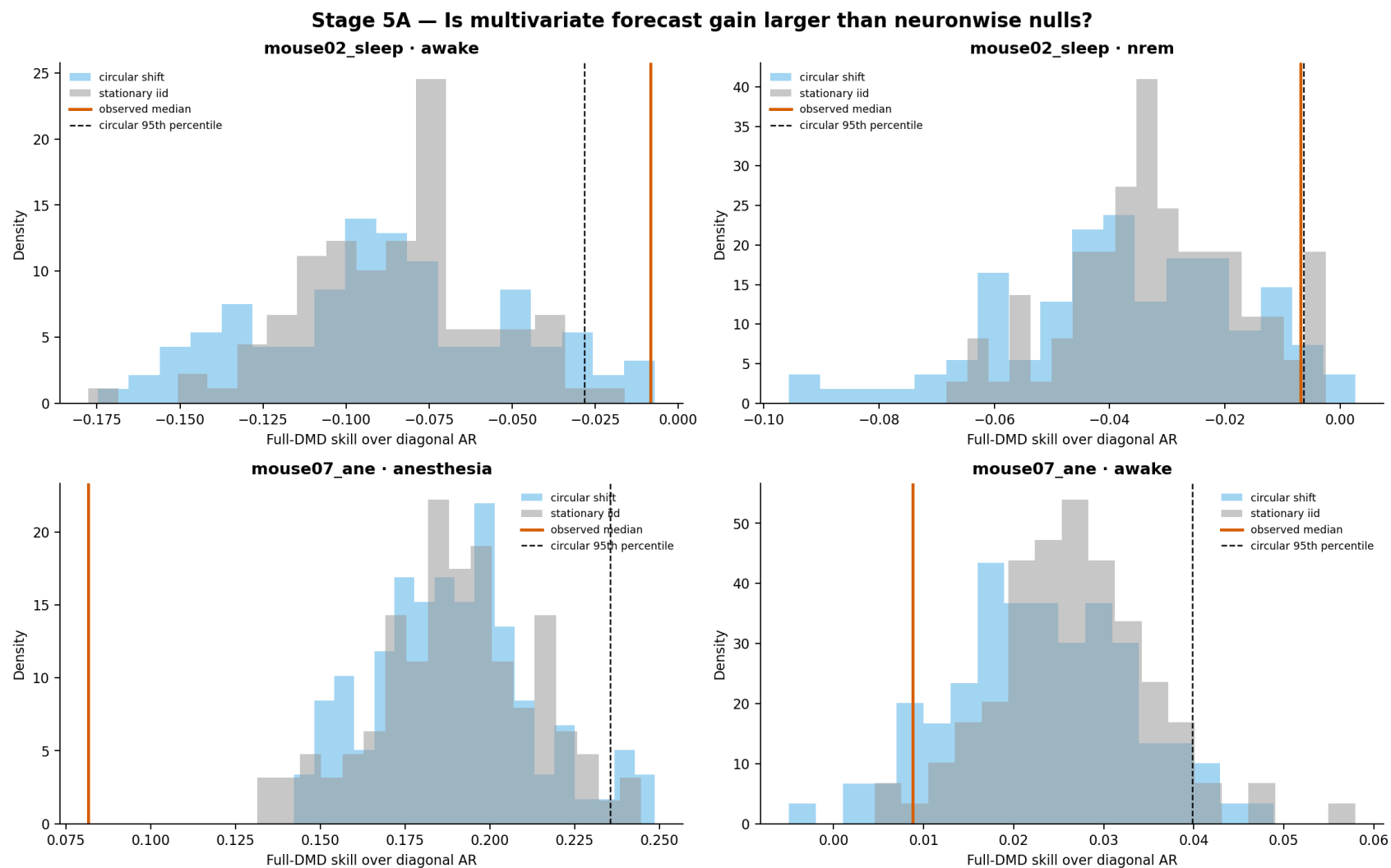


Figure 20: Observed full-over-diagonal skill (orange) against synchronized cell-level circular-shift and stationary-null distributions.

Null subspace similarity is also comparable to the already-low observed value, especially in mouse02. Neither observed nor null distributions approach the Gate-4 threshold.

12 Final gate application

Gate	Frozen threshold	Observed	Result
1. Geometry / numerics	Zero illegal joins; at least 95% finite	Zero illegal geometry; 100% finite; zero bootstrap joins; zero explosions	PASS
2. Known systems	Class ≥ 0.80 ; $S_{\text{sub}} \geq 0.75$; eigenvalue error ≤ 0.25 ; skill > 0	0.55; 0.430; 0.160; 0.324	FAIL
3. Prediction	Lower bound > 0 in at least 3/4 cells; no 2-s failure; $< 5\%$ explosive	4/4; 4/4 positive at 2 s; 0% explosive	PASS
4. Reproducibility	Median $S_{\text{sub}} \geq 0.80$ in at least 3/4; subset class agreement	0/4; subset classes agree 4/4 but matching fails	FAIL
5. Resolution	$R_{\text{track}} < 0.5$ in both recordings	0.819 (ineligible) and 0.854	FAIL

The frozen rule assigns **FAIL** if sparse/calcium-like known-system recovery fails, even when empirical prediction passes. Gates 4 and 5 independently reach the same practical conclusion. The coordinated-mode qualifier fails in all four cells.

Authorized conclusion

The data contain forecastable temporal structure, but this pilot cannot identify or track reproducible DMD modes at the proposed 300-frame resolution. Do not claim awake/NREM/anaesthesia differences in DMD eigenvalues or subspaces from this implementation, and do not proceed to intensive Grassmann tracking without a newly reviewed measurement/window model.

13 Limitations and defensible next options

13.1 What this pilot does not prove

- It does not prove that the underlying neural dynamics are non-linear or lack low-dimensional structure; it shows that the present observation/window pipeline cannot recover that structure reliably.
- It does not compare brain states statistically. Only two recordings and three fixed evaluation windows per cell were used.
- Bootstrap intervals condition on the selected windows and PCA basis. They quantify conditional operator-estimation perturbations, not uncertainty from test origins, window/neuron selection, preprocessing, model selection, animals, or experiments.
- The 1,500-frame diagnostic reselects ranks up to 32 and cannot isolate window length from model capacity.
- Stored smoothing can borrow support near label boundaries; the fixed edge trim reveals sensitivity but does not reconstruct the upstream smoother.
- Circular shifts use periodic wrap inside a legal bout. No shifted sample leaves the state/acquisition bout, but the wrap is a known surrogate artefact.

13.2 If the project is revisited

The most defensible new work would begin with a separate, prospectively frozen plan and one of the following endpoints:

1. a calcium-aware latent state-space/switching model fitted to fluorescence or deconvolution uncertainty, with simulation-calibrated identifiability;
2. coarser event-count/rate observations and much longer windows, with the affine mean and diagonal baselines retained as primary comparators;
3. static or slowly varying rate/covariance summaries if the scientific aim is brain-state characterization rather than modal dynamics;
4. if modal tracking is retained, a prospectively eligible slow/predictive spectral cluster represented by ordered real-Schur subspaces, allowing real-complex transitions without post-hoc mode re-ranking; or
5. additional recordings and animal-level validation before any state effect claim.

The P arm may remain useful for a prediction benchmark, but the present results do not justify using its eigenvectors as biological objects.

14 Reproducibility and artifact inventory

The final run contains 109 files (approximately 40 MB) across eight completed stages. Generated results remain inside the private DMD/ tree. The accepted implementation-tree SHA-256 is:

```
95ee656eb4c9057fc004b76aebcd670418826be37bc620fc7afb470ebdca48f0
```

It matches the current 24 hashed files under DMD/src, DMD/scripts, DMD/tests, and DMD/configs. The frozen configuration hash is stored in config_frozen.json; the Git revision is:

```
ca0ddd9670b64bfbe8fd3565d005cb324642964d
```

Because the DMD files are currently untracked, content hashes—not the Git revision alone—identify the implementation.

Verification commands:

```
PYTHONPATH=DMD/src poetry run python -m unittest discover -s DMD/tests -v
poetry run ruff check DMD/src DMD/scripts DMD/tests
```

All 25 tests pass and Ruff reports no violations. Tests cover selective HDF5 orientation, window legality, deterministic maximin placement, per-window B4, frozen RMS scaling/PCA, known linear recovery, ridge scale invariance, exact forecast powers, no cross-segment pairs, conjugate grouping, rejection of full-space projectors, projector geometry, fixed-reference matching, no bootstrap-block joins, within-bout circular shifts, stationary null sampling, fixed split preservation, independent long-block diagonal selection, and synchronized cell summaries.

Primary machine-readable entry points are:

- active_run.json — accepted run pointer;
- stage_status.json — timestamps and stage summaries;
- software_manifest.json — versions and per-file hashes;
- selection_manifest.json — development-only winners;
- predictive_bootstrap_cell_summary.csv;
- tracking_stability_cell_summary.csv;
- tracking_resolution_summary.csv;
- negative_control_cell_summary.csv; and
- final_decision.json.

A Additional PC-trace diagnostics

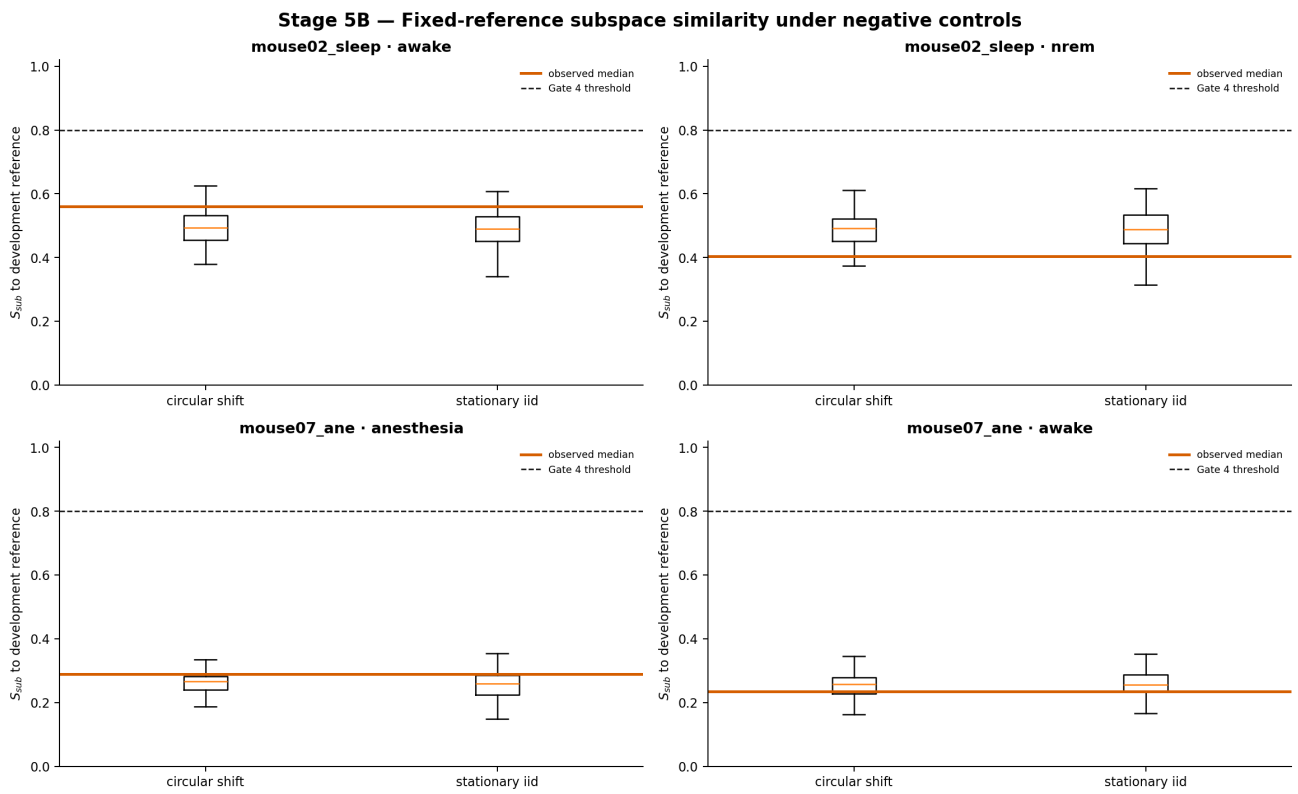


Figure 21: Similarity to the fixed development reference under circular and stationary controls. Orange lines are original evaluation medians; dashed lines mark 0.80.

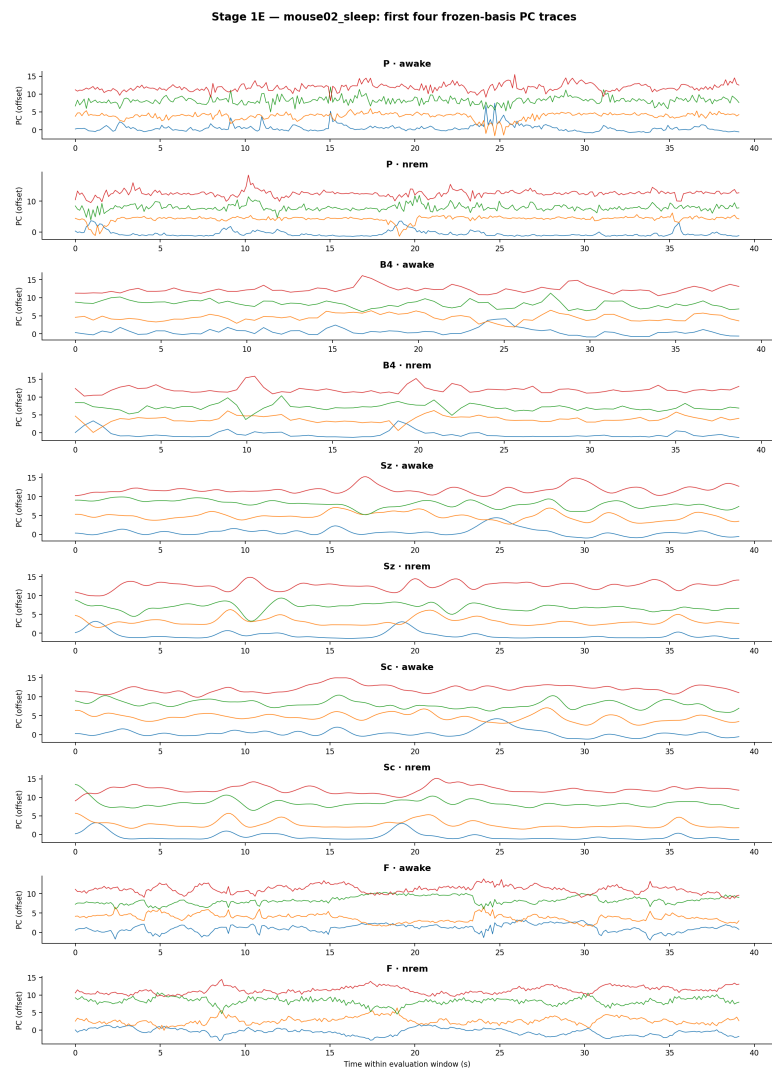


Figure 22: Mouse02 first four frozen-basis PC traces for every arm and state.

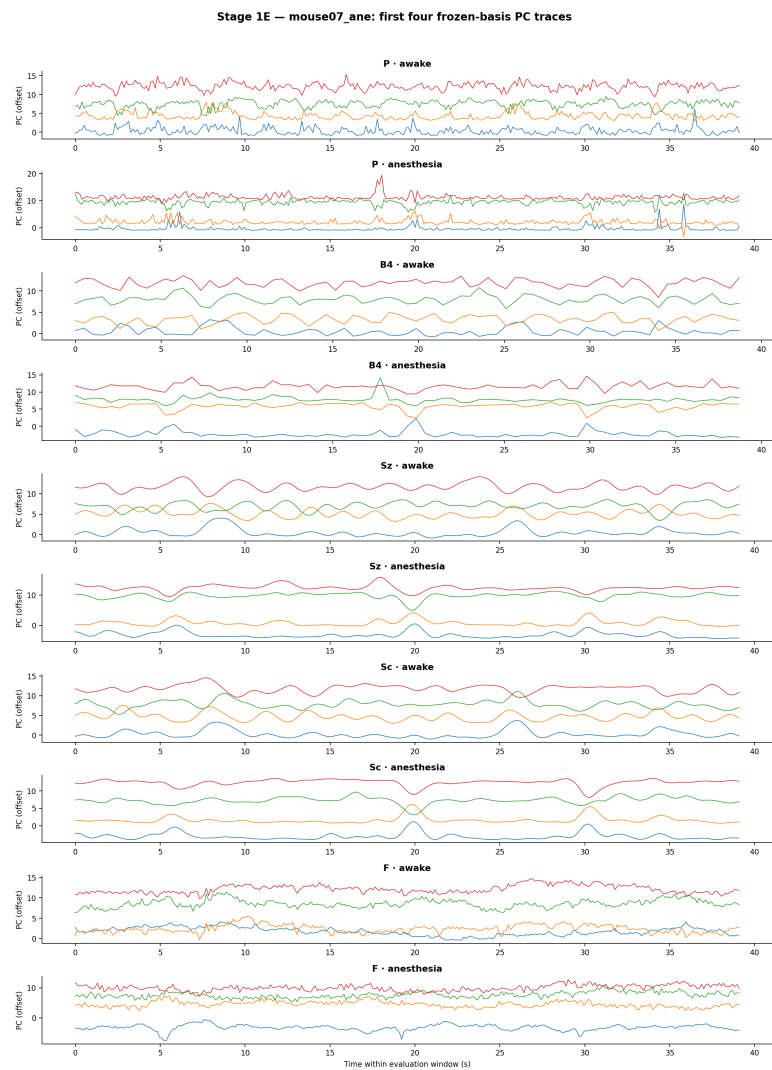


Figure 23: Mouse07 first four frozen-basis PC traces for every arm and state.

B Exact selected window intervals

All intervals below are zero-based and half-open:

Recording	Label	Chronological 300-frame windows
mouse02	awake	D [302, 602); E [2881, 3181); D [5962, 6262); E [9350, 9650); D [11929, 12229); E [14508, 14808)
mouse02	NREM	D [961, 1261); E [3897, 4197); D [7178, 7478); E [10270, 10570); D [15464, 15764); E [18400, 18700)
mouse07	awake	D [0, 300); E [1840, 2140); D [3680, 3980); E [5520, 5820); D [7360, 7660); E [9200, 9500)
mouse07	anaesthesia	D [9500, 9800); E [11340, 11640); D [13180, 13480); E [15020, 15320); D [16860, 17160); E [18700, 19000)

Long blocks are mouse02 awake [12465, 13965), mouse02 NREM [16918, 18418), mouse07 awake [4000, 5500), and mouse07 anaesthesia [13500, 15000).

References

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